

Class 18 Mini Project

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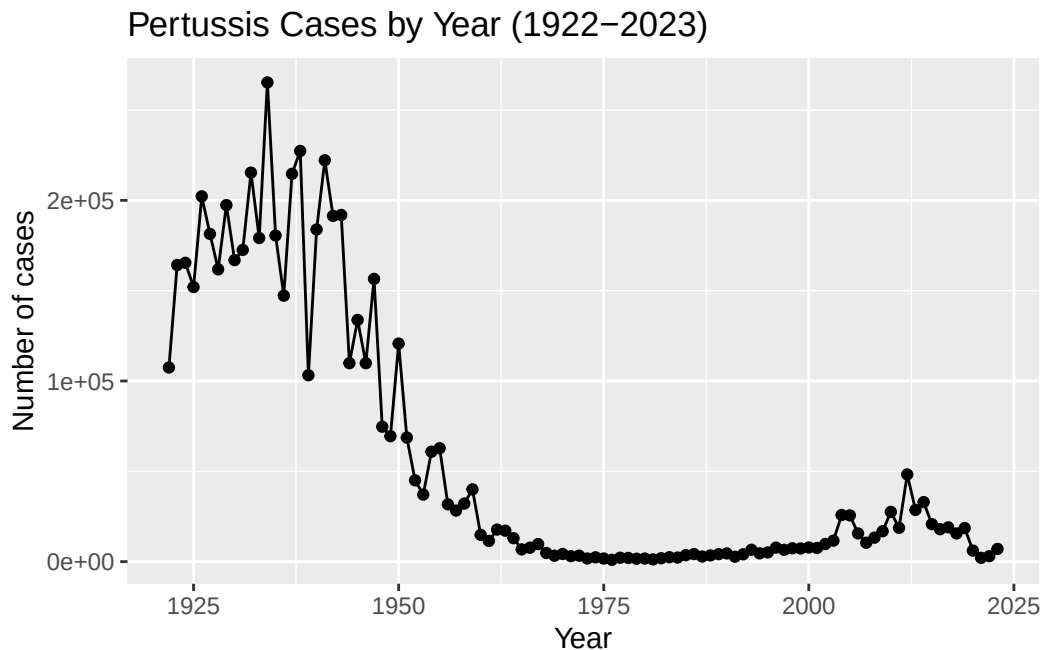
Investigating pertussis cases by year

First we have to install the **datapasta** package.

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
# install.packages("datapasta")
library (datapasta)
library(ggplot2)
cdc <- data.frame(
  year = 1922:2023,
  cases = c(
    107473, 164191, 165418, 152003, 202210, 181411, 161799, 197371,
    166914, 172559, 215343, 179135, 265269, 180518, 147237, 214652,
    227319, 103188, 183866, 222202, 191383, 191890, 109873, 133792,
    109860, 156517, 74715, 69479, 120718, 68687, 45030, 37129,
    60886, 62786, 31732, 28295, 32148, 40005, 14809, 11468,
    17749, 17135, 13005, 6799, 7717, 9718, 4810, 3285,
    4249, 3036, 3287, 1759, 2402, 1738, 1010, 2177,
    2063, 1623, 1730, 1248, 1895, 2463, 2276, 3589,
    4195, 2823, 3450, 4157, 4570, 2719, 4083, 6586,
    4617, 5137, 7796, 6564, 7405, 7298, 7867, 7580,
    9771, 11647, 25827, 25616, 15632, 10454, 13278, 16858,
    27550, 18719, 48277, 28639, 32971, 20762, 17972, 18975,
    15609, 18617, 6124, 2116, 3044, 7063
  )
)
ggplot(cdc) +
  aes(x=year, y=cases) +
  geom_point() +
```

```
geom_line() +
labs(x="Year", y="Number of cases", title= "Pertussis Cases by Year (1922-2023)" )
```



A tale of two vaccines (wP & aP)

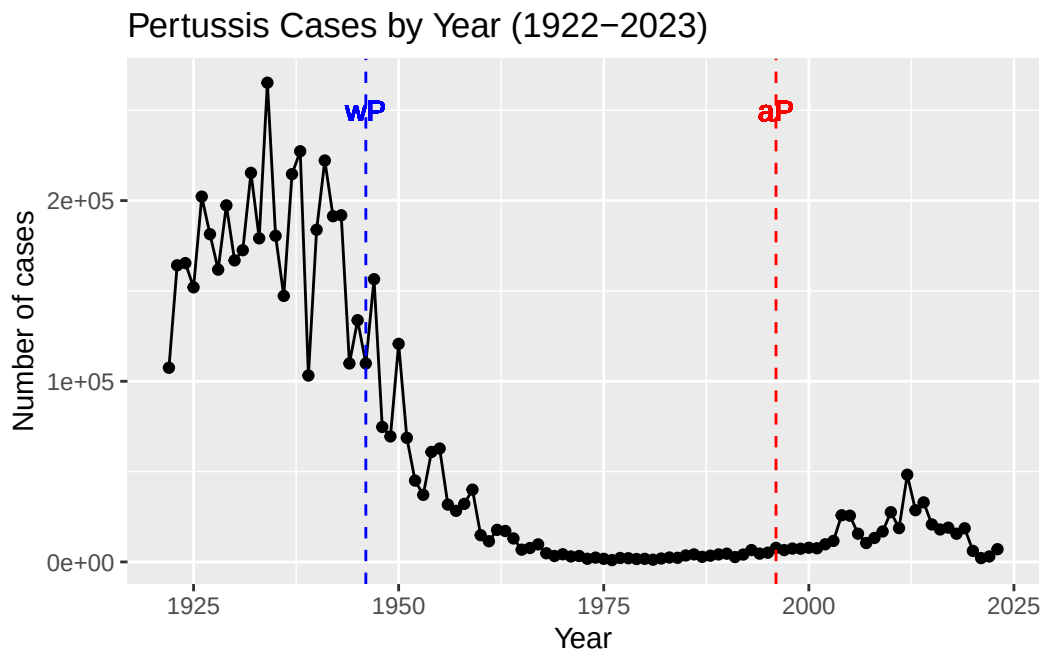
Examining the CDC data plot to see the changes that happened after the switch to the acellular pertussis (aP) vaccination program

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x=year, y=cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, color = "blue", linetype= "dashed") +
  geom_vline(xintercept = 1996, color = "red", linetype= "dashed") +
  geom_text(aes(x=1946, y=250000, label="wP"),color="blue") +
  geom_text(aes(x=1996, y=250000, label="aP"), color="red") +
  labs(x="Year", y="Number of cases", title= "Pertussis Cases by Year (1922-2023)" )
```

Warning in `geom_text(aes(x = 1946, y = 250000, label = "wP"), color = "blue")`: All aesthetics must be placed in the `aes()` function.
Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_text(aes(x = 1996, y = 250000, label = "aP"), color = "red")`: All aesthetics must be placed in the `aes()` function.
Please consider using ``annotate()`` or provide this layer with data containing a single row.



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

- After the introduction of the aP vaccine, the number of Pertussis cases started to increase over time (as compared to the decreased/low cases before introduction). Even though the latter aP vaccines were developed to have less side effects, since they used only purified antigens of the bacteria instead of the broader ‘whole cell’ inactivated bacteria, the immune response from the aP vaccine may be more narrow/less lasting as compared to the earlier vaccine, leading to an increase in the number of cases. Other hypotheses include more sensitive PCR-based testing, vaccination hesitancy, bacterial evolution (escape from vaccine immunity) and waning of the immunity in adolescents originally primed as infants with new aP vaccine as compared to the older wP vaccine.

Exploring CMI-PB data

The CMI-PB API returns JSON data

For reading such files, we need to use the `read_json()` from the **jsonlite** package.

```
library(jsonlite)
```

Read the main subject database table from the CMI-PB API.

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

- There are 87 aP and 85 wP infancy vaccinated subjects in the dataset.

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female  Male  
112     60
```

- There are 112 female and 60 male subjects in the dataset.

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

	American Indian/Alaska Native	Asian	Black or African American
Female	0	32	2
Male	1	12	3

	More Than One Race	Native Hawaiian or Other Pacific Islander
Female	15	1
Male	4	1

	Unknown or Not Reported	White
Female	14	48
Male	7	32

Working with dates

We use the **lubridate** package.

```
# install.packages("lubridate")
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

Today's date

```
today()
```

```
[1] "2026-03-06"
```

How many days have passed since new year 2000?

```
today() - ymd("2000-01-01")
```

Time difference of 9561 days

What is this in years?

```
time_length( today() - ymd("2000-01-01"), "years")
```

```
[1] 26.17659
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(lubridate)

subject$age <- today() - ymd(subject$year_of_birth)

library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
ap <- subject %>% filter(infancy_vac == "aP")

round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Q8. Determine the age of all individuals at time of boost?

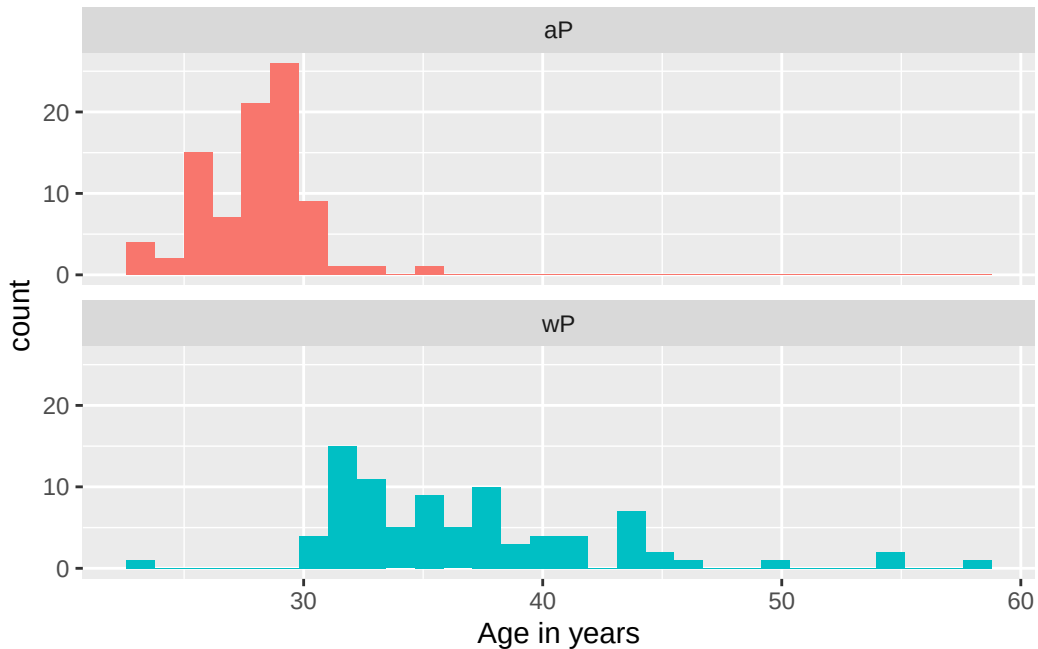
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
library(ggplot2)
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



p-value

```
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

- From the plot and the p-value, it can be concluded that the two groups are statistically significantly different.

Joining multiple tables

```
# Complete the API URLs...
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:


```
meta <- left_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost			
1	1	1	-3			
2	2	1	1			
3	3	1	3			
4	4	1	7			
5	5	1	11			
6	6	1	32			

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	14674 days
2	14674 days
3	14674 days
4	14674 days
5	14674 days
6	14674 days

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4  
6698  5389 10117 10124 10124 10124
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset  
      31520           8085           7301           5670
```

- The most recent dataset (2023 dataset) has the least number of rows (5670), lower than the past few years- it has the least data.

Examine IgG Ab titer levels

```
igg <- abdata %>% filter(isotype == "IgG")  
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

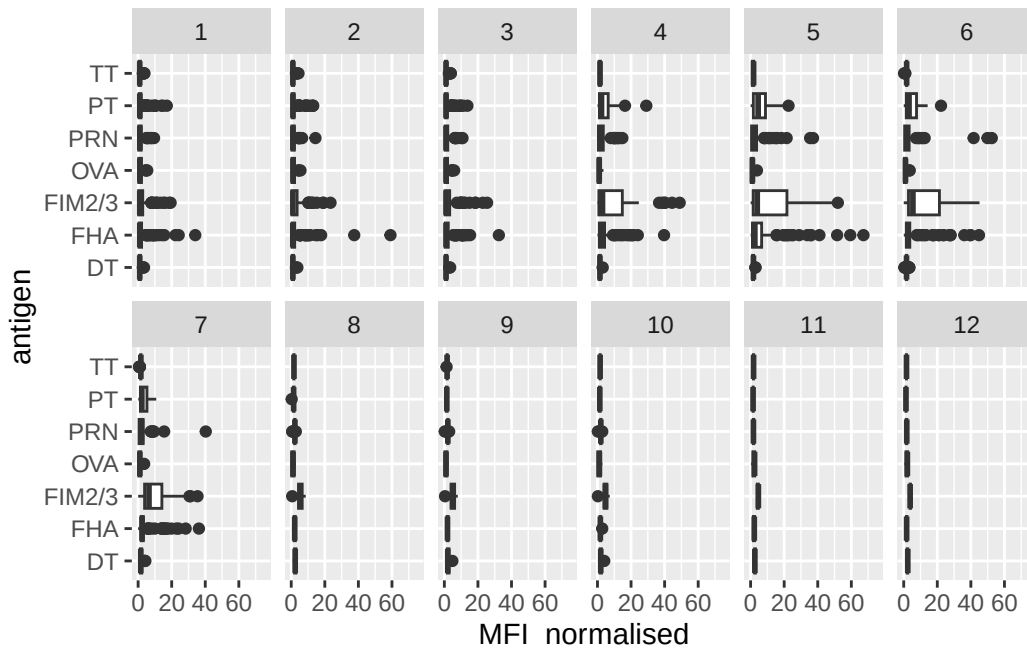
	age
1	14674 days
2	14674 days
3	14674 days
4	15770 days
5	15770 days
6	15770 days

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
```

```
xlim(0,75) +
facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

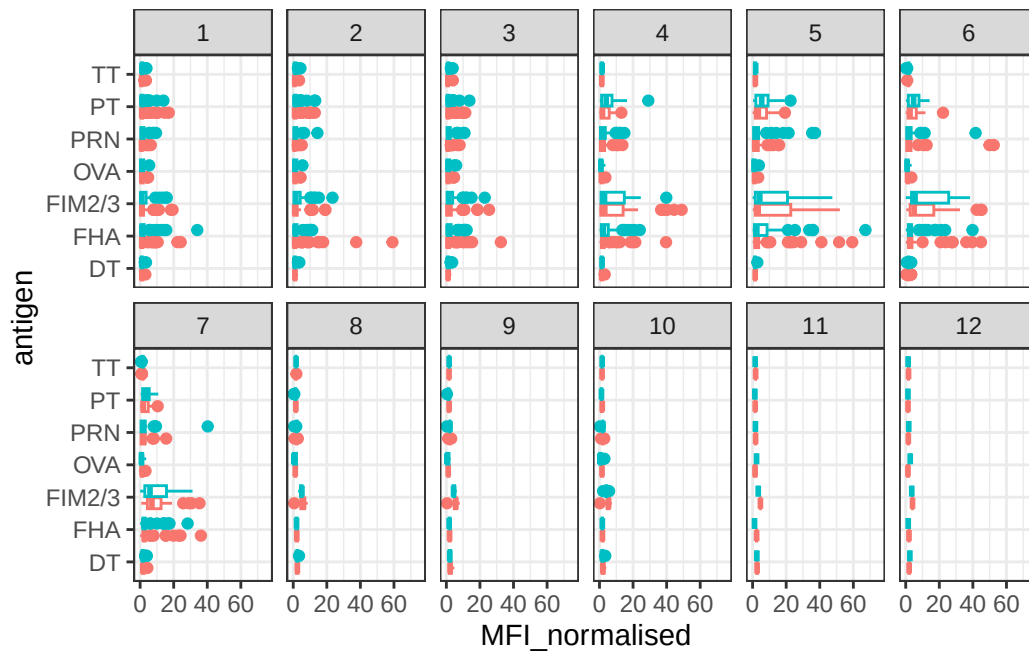


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

- PT, PRN, FIM2/3 and FHA show differences in the level of IgG antibody titers recognizing them over time. These are pertussis-related antigens which are targets of the vaccine/booster response therefore they are expected to be strongly boosted after vaccination. Others are not pertussis-related antigens and therefore are not expected to be strongly boosted after vaccination.
- setting color and/or facet values of the plot to include `infancy_vac` status:

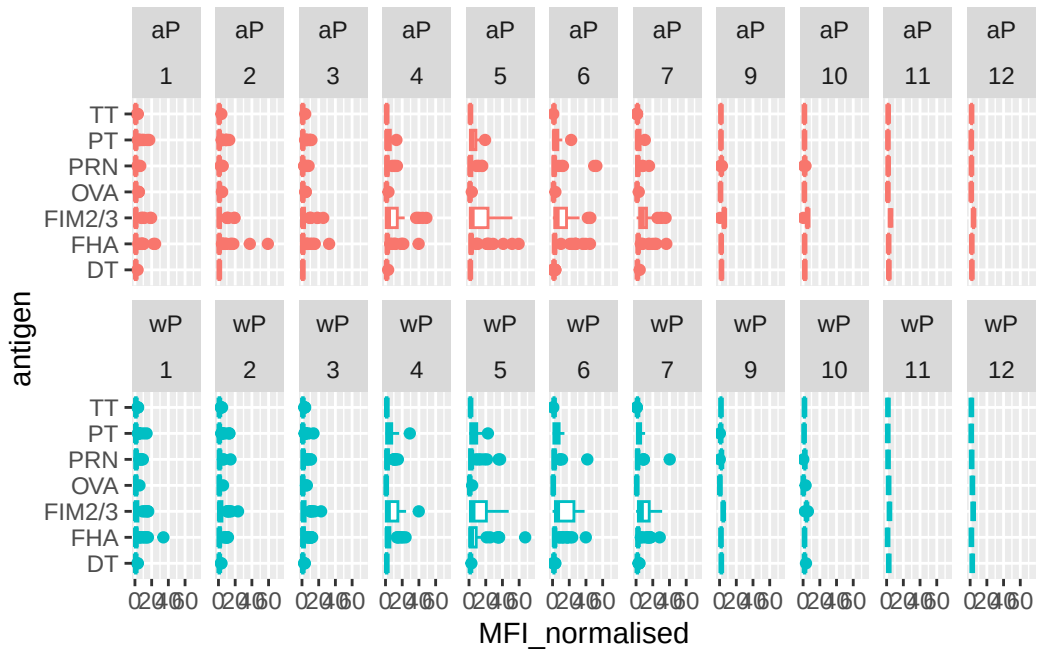
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range
(`stat\_boxplot()`).



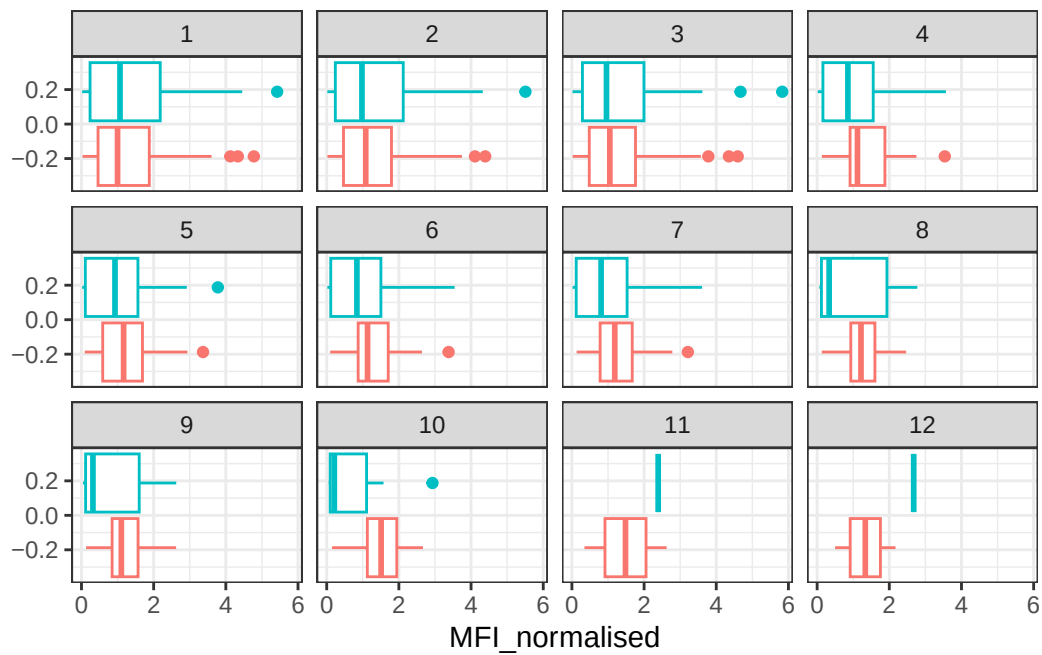
```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range
(`stat\_boxplot()`).

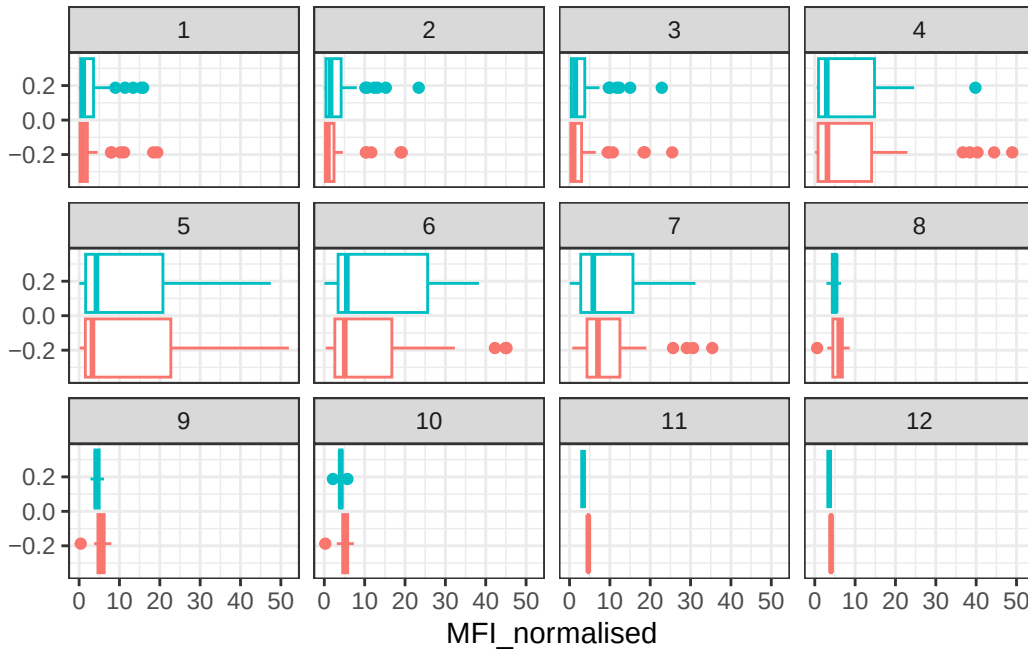


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

- PT levels increase over time and clearly exceed the OVA levels. Their peak is seen around visit 5 and then they decline, this is similar for both the wP and aP subjects.

Q17. Do you see any clear difference in aP vs. wP responses?

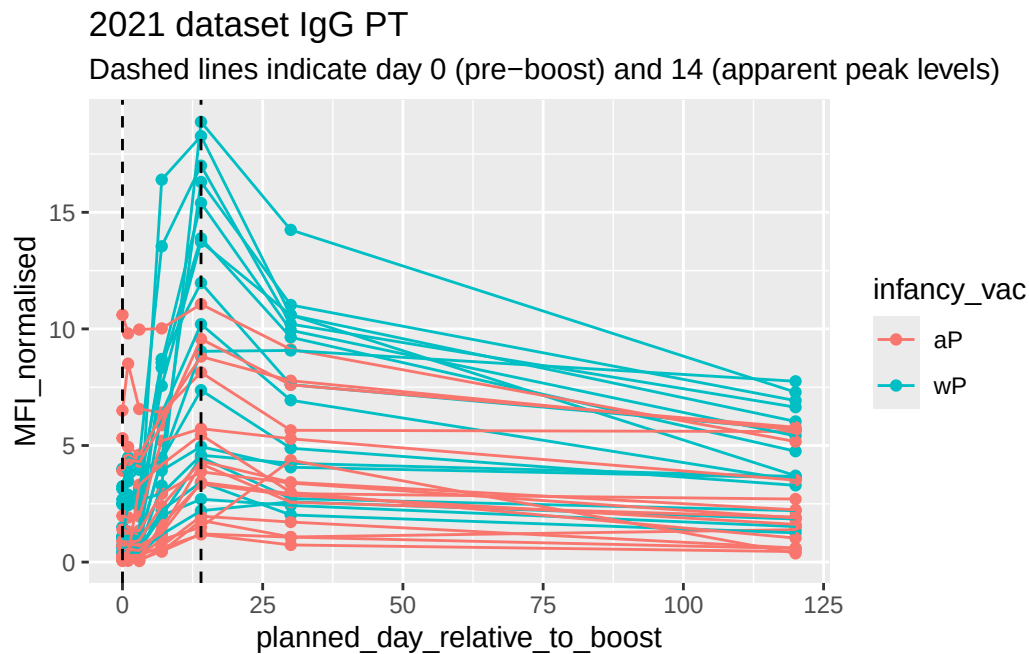
- No, there are no major clear difference in aP vs. wP responses. Both groups show similar overall patterns with PT levels increasing after the boost and then declining afterwards.
- Looking at the 2021 dataset IgG PT antigen levels over-time:

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
  geom_point() +
  geom_line() +
```



```
geom_vline(xintercept=0, linetype="dashed") +
geom_vline(xintercept=14, linetype="dashed") +
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



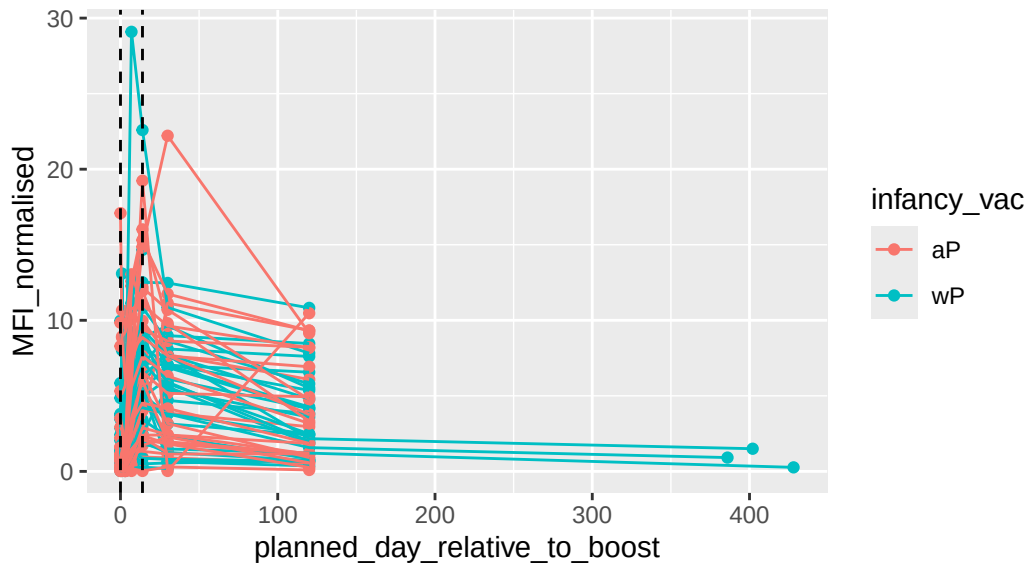
Q18. Does this trend look similar for the 2020 dataset?

```
abdata.21 <- abdata %>% filter(dataset == "2020_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2020 dataset IgG PT",
          subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2020 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



- Yes, the trend looks similar overall as the IgG PT levels increase after the boost and then decline afterwards but it should be noted that between the 2021 and 2020 datasets- 2020 data is noisier and appears more scattered so the trend is less clean as compared to the cleaner 2021 data set.

Obtaining CMI-PB RNASeq data

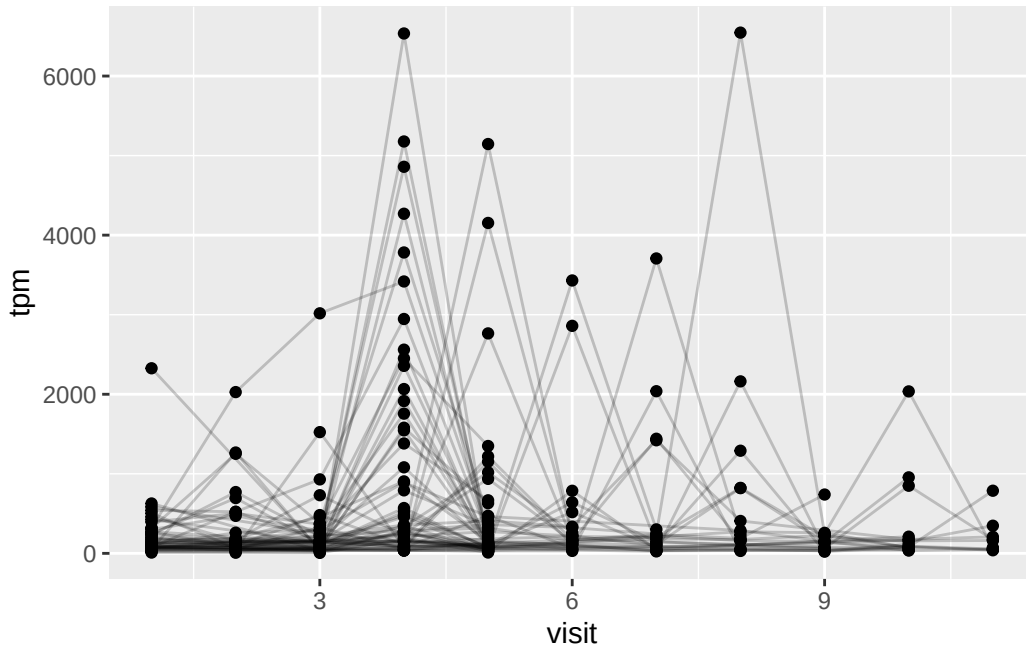
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)

ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



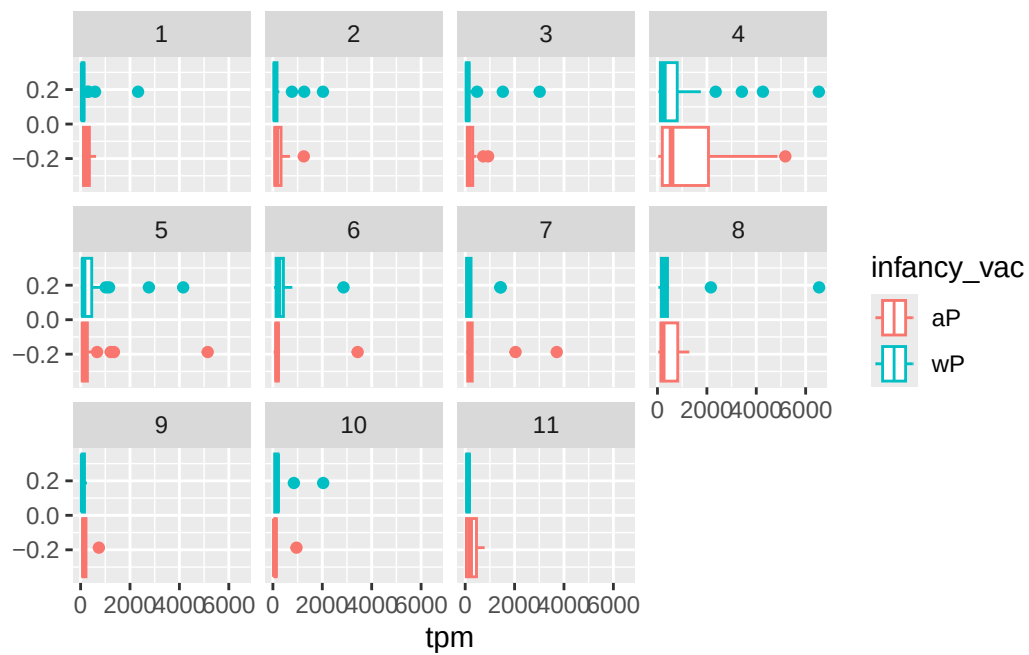
Q20.: What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

- The expression of this gene is at its maximum level around visit 4 (lower at earlier visits) and then it decreases again after visit 4.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

- This pattern does not exactly match the antibody titer data which showed longer elevation while this patterns shows that it peaks around visit 4 and then declines afterwards. This may be possibly due to the fact that antibody levels can stay remain even after gene expression has declined therefore we see elevated titer data levels while this gene expression shows earlier decline.

```
ggplot(ssrna) +
  aes(tpm, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(vars(visit))
```



```
ssrna %>%
  filter(visit==4) %>%
  ggplot() +
    aes(tpm, col=infancy_vac) + geom_density() +
    geom_rug()
```

